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Florida Hurricane Preparedness Algorithm

I. Introduction

Hurricanes are among the most devastating natural disasters, causing widespread destruction to infrastructure while leaving thousands displaced. Florida, with its extensive coastline, is particularly vulnerable to hurricanes, making it imperative to identify high risk areas to ensure efficient resource allocation and emergency planning. Our project addresses this challenge by developing an algorithm that ranks Florida counties based on their vulnerability during a hurricane. The algorithm incorporates data from historical hurricanes, population density, and energy consumption to determine the most at-risk areas.

This report outlines our approach to ranking high-risk counties, focusing on three critical parameters: the path of the hurricane, population density, and electrical grid vulnerability. By combining these datasets and applying a weighted scoring system, our model provides actionable insights that can assist local governments and emergency response teams in prioritizing aid and resources.

II. Dataset Description

Hurricane Path Data

The hurricane path data includes the historical tracks of three significant hurricanes: Hurricane Fay, Hurricane Idalia, and Hurricane Irma. Each of these datasets record the coordinates of the hurricane's path, its wind speed, and its Saffir-Simpson category. These attributes are essential for assessing the intensity and geographical impact of each storm on Florida's counties. The data was sourced from NOAA's Historical Hurricane Tracks repository, ensuring reliability and consistency.

Population Density Data

The population density data provides insights into the number of people residing in each county. Higher population density signifies greater potential for human impact during a hurricane. This dataset is crucial for understanding the social vulnerability of each region and was obtained from the 2021 U.S. Census.

County Location Data

To determine the distance from the hurricane's path to each county, a central coordinate was picked from each of the 67 counties to approximate how far the historical hurricanes were from each county.

Electrical Grid Data

To understand the risk to electrical grid infrastructure, we analyzed the capacity and generation of electricity across all the Florida counties. This dataset, sourced from the U.S. Energy Information Administration (EIA), includes critical parameters such as total energy generated, and peak load demands for each county. This provides a comprehensive view of the energy allocation in Florida, showing patterns of high energy consumption and capacity which can be potential stress points on the grid.

Regions with higher energy consumption were flagged as higher-risk areas, as their elevated demand increases the likelihood of grid failures during hurricanes. High consumption can strain infrastructure, especially under the added pressure of extreme weather conditions, making these areas particularly susceptible to prolonged disruptions. This data-driven approach allowed for a quantifiable assessment of risk, enabling the identification of counties where infrastructure improvements or preventative measures are most urgently needed.

By focusing on consumption and generation trends, the analysis emphasizes the importance of proactive planning to fortify Florida's energy systems. Insights gained from this study can inform strategies to enhance grid resilience, allocate resources effectively, and reduce vulnerability in the face of increasingly severe weather events.

Integration and Analysis

The four datasets were normalized and combined into a single weighted scoring model. Each parameter—hurricane path proximity, county location, population density, and electrical grid vulnerability, hurricane risk index, and national risk index—was assigned a weight based on its significance in determining hurricane risk. The weighted scores were then used to rank counties, highlighting the areas that are most in need of evacuation planning, emergency supplies, and power restoration efforts.

At first, the data was collected to only analyze the three parameters: hurricane path, population density, and electrical grid vulnerability. But after implementing the data, there was an issue to address: to compare the hurricane's latitude and longitude with an unknown choice of regions in the state of Florida. Additionally, new data was collected where a hurricane risk index and national disaster risk index was found, so this was also integrated into the algorithm. For this

project, there was data collected on a coordinate for each of the 67 counties in Florida, as a tether for the hurricane's path to compare with. When considering the distance between the hurricane and the counties, another consideration was determining how to calculate this distance. To account for the curvature of the earth, since the coordinates are based in degrees originally, it made most sense to use the Haversine formula to calculate the distance between the hurricane and each county. Then there is data for the wind speed and hurricanes.

Overall, a covariance matrix is computed using the normalized data and from it, the eigenvectors are found. Then the eigenvector with the largest eigenvalue is used for the weights for each category that influences each county, such as population. The risk score is determined from a weighted average based on the weights calculated.

IV. Applied Algorithms

For this algorithm, the risk factors of hurricane proximity, hurricane category, population, and energy capacity are loaded from data files into datasets, and then are weighted using Principal Component Analysis (PCA) and then implementing a weighted-score algorithm to rank counties most affected.

When calculating the distance between a hurricane's location and then a coordinate from each county, the Haversine formula was implemented. Since the distance between these two locations is not like on a flat Euclidean plane, the Euclidean distance is not sufficient. To account for the curvature of the Earth, the Haversine formula is used.

The assumption is that Earth is modeled as a sphere in this code, so the radius of the sphere is designated as a constant, R , as 6371 km.

Then using the latitude and longitude coordinates of the data from the counties and hurricanes, we can calculate the distance between these components, which as follows:

Difference in latitude.

$$\Delta\phi = \phi_2 - \phi_1$$

Difference in longitude.

$$\Delta\lambda = \lambda_2 - \lambda_1$$

Using these values, then we can compute the central angle with a rendition of the haversine function based on trigonometry.

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos\phi_1\cos\phi_2\sin^2\left(\frac{\Delta\lambda}{2}\right)$$

The central angle to compute the great-circle distance between two points on a sphere is c .

$$c = 2 \arctan\left(\frac{\sqrt{a}}{\sqrt{1-a}}\right)$$

Then distance between the two points is finally calculated using

$$R \cdot c$$

which is returned as a value to find the hurricane proximity score. Along with calculating the distance between the hurricane and each county, there was also a consideration for the hurricane categories in the historical data, dependent on the wind speed in the unit, knots (kt). For example, if a hurricane had wind speeds of 157 kt or above, it is considered a Category V hurricane according to the Saffir-Simpson scale.

The Hurricane proximity calculation takes the proximity of each county to the nearest hurricane using the Haversine formula. Then it finds the minimum distance between each county and the storms, and then adds it into the hurricane proximity score.

Then to score the data effectively and to make sure that the variables were all comparable despite having different units of measurements (i.e. knots and degrees), this was accomplished using min-max normalization. This takes each data point and scores it from 0 to 1, also known as normalization. The formula for this is as follows:

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Where x is each data point and x' is the normalized data point. This is compared to the largest and smallest data points in each dataset. The purpose of this normalization is to compare the values and rank based on the assigned weights.

Then the next step was to derive the weights for each parameter based on the variance of the data, which is a litmus for their contribution to the data. The overall method to find the counties most at risk was with Principal Component Analysis (PCA). A covariance matrix was used to find patterns in the variance between the data and then find the principal components, or in other words, find the most variance.

The covariance between two variables is calculated by:

$$Cov(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

Where $\bar{x}, \bar{y}$ are the sample means of x and y and n is the number of data points. Then the the covariance matrix for three parameters: x, y, z would look like:

$$C = \begin{bmatrix} Cov(x, x) & Cov(x, y) & Cov(x, z) \\ Cov(y, x) & Cov(y, y) & Cov(y, z) \\ Cov(z, x) & Cov(z, y) & Cov(z, z) \end{bmatrix}$$

Where the matrix's diagonal elements represent the variance of each variable onto itself. Then there is commutativity, since there is symmetry along this diagonal. The three parameters are the hurricane's path, the population, and then energy consumption data points.

The resultant matrix would show covariance, or how much variance there is in the data between the parameters. Once the matrix is made, then to finish the Principal Component Analysis, the eigenvalues and eigenvectors are determined from this matrix. Positive values represent an increasing relationship, while negative values indicate the inverse.

Then once the covariance matrix is populated, the eigenvalues and then eigenvectors are calculated. The eigenvector determines the direction of the variables, and the eigenvalues indicate how much variance there is in each vector. To find the eigenvalues, we can write this:

$$C\vec{v} = \lambda\vec{v}$$

Where C is the covariance matrix, $\vec{v}$ is the eigenvector, and λ is the eigenvalue. Using Linear Algebra, this can be found from a variation of the equation:

$$(C\vec{v} - \lambda I)\vec{v} = 0$$

Then this gives us:

$$\det(C - \lambda I) = 0$$

The determinant of the matrix is a characteristic polynomial, and its solutions are the eigenvalues. Once the eigenvalues are found, then each eigenvector can be solved by substituting

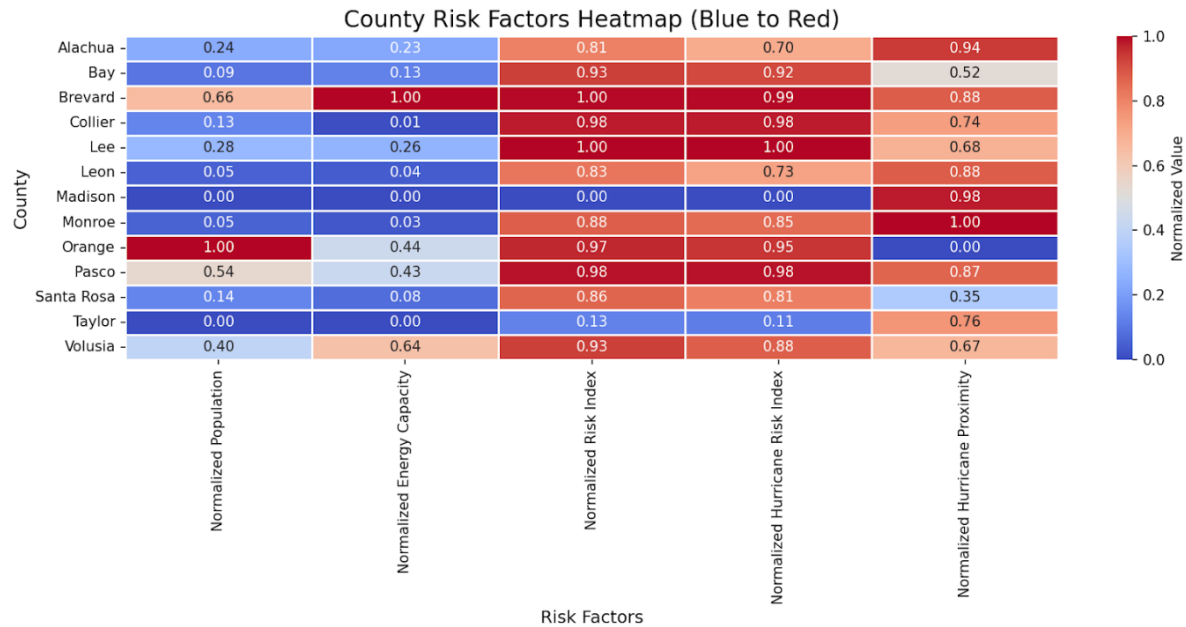
each λ back into the equation and then solving for v . This process is repeated until all the variables are accounted for.

The eigenvector with the largest eigenvalue indicates the most variance from the covariance matrix, so hence, determines what variables have impacted the data the most. That is why the max eigenvector is used in the code calculation. Once the eigenvector has been identified, then the weights are normalized, which its entries are the weights used in the weighted-scoring algorithm.

Then with the calculation of the weights, the risk score is determined by a weighted sum, taking each county and multiplying by the weights calculated from PCA. This effectively assigns values to each county and determines overall hurricane risk. After a weighted risk score is assigned for each county, then the counties are sorted by their risk scores. The higher the risk score, the higher the risk. Then a heatmap is plotted to visualize the normalized risk factors scored from 0 to 1.

V. Results

Our algorithm ultimately results in a heatmap, where all the normalized parameters are on a scale from 0 to 1 (blue to red). Each parameter is shown with its own risk score for every county in the path of the hurricane. The counties that are at least at risk of being damaged by a hurricane have risk scores closer to 0 (or bluer), while the most at-risk counties have more risk scores closer to 1 (or redder). The results and the inclusion of the normalized parameters allow us to decide which counties are the most at-risk when the scores are tied, and what these at-risk counties may need more in terms of support.



```

Risk scores calculated:
  County Risk Score
0 Alachua 0.575290
3 Bay 0.566310
5 Brevard 0.913918
7 Collier 0.603914
9 Lee 0.685497
Ranked counties:
  County Risk Score
5 Brevard 0.913918
20 Pasco 0.780608
19 Orange 0.765162
24 Volusia 0.728052
9 Lee 0.685497
Ranked data has been saved to: ranked_hurricane_risk.csv
Final ranked results:
  County Total Capacity (MW) Unique Technologies ... Normalized Risk Index Normalized Hurricane Proximity Risk Score
5 Brevard 20886.6 5.0 ... 0.995539 0.879297 0.913918
20 Pasco 9047.0 4.0 ... 0.984900 0.866023 0.780608
19 Orange 9168.0 4.0 ... 0.968428 0.000000 0.765162
24 Volusia 13297.6 4.0 ... 0.932395 0.671163 0.728052
9 Lee 5479.8 4.0 ... 1.000000 0.684475 0.685497

```

The above figure depicts the standard output when running our algorithm. It presents the risk scores of each county and sorts the top counties based on how close their risk scores are to 1. In the final ranked results, the risk score at the far right determines how at risk a county is compared to the others. For example: Brevard has an overall risk score of 0.9 according to our algorithm, meaning it will need the most support during the event of a hurricane, compared to the

counties of Pasco, Orange, Volusia, and Lee. These final rankings are saved to a separate csv file.

Running Time Analysis

After analyzing our algorithm and comparing it to third-party hurricane risk tracker tools, our algorithm is more specific yet has the same accuracy as other hurricane risk tracker tools. Counties of Florida located along the coastline are at a higher risk than counties inland. This is the same for Online risk tracker tools such as Florida's Know Your Zone map or FEMA's National Risk Index map. However, our algorithm also individually keeps track of data that may or may not be already combined with hurricane risk tracker tools, such as population and energy capacity. At the same time, our algorithm takes longer than online hurricane risk tracker tools because its calculations require real-time information to be an effective tool.

Hurricanes are unpredictable, and their paths can move in unpredictable ways. Florida's Know Your Zone map or FEMA's National Risk Index map can pull up a risk score within moments. However, they do not consider what areas may be affected by the hurricane, so they process values very quickly. Since our algorithm needs to consider where the hurricane is coming from and where it's going to, it takes much longer for our algorithm to find the area and normalize the risk scores when the data is needed now.

The runtime will change based on the amount of information given during the event of a hurricane, and for each function, it may differ based on how many counties, or other variables, are included:

File Loading: $O(n)$

The file loading function takes $O(n)$, where n is the number of strings and integers read from the file. The value of n changes depending on how much information is provided. For hurricane path

files, the value of n will vary based on how many locations of the hurricane's path have been recorded. For the files recording energy consumption and risk index, the value of n remains the same unless those parameters become outdated.

Data Merging: $O(n \log n)$

The data merging function takes $O(n \log n)$ in a similar way that MergeSort does. First, all datasets are combined to keep everything in a single place. The merged data would include the energy capacity and population data, the risk scores (national and hurricane risk scores), and the path of the given hurricane(s). Then, after each parameter is normalized, they are merged again with the normalized values to be used in calculating their weights. The value of n depends on the number of datasets needed to be analyzed and normalized during the merging and normalization process. With multiple datasets being merged, the overall time complexity is $O(n \log n)$ where n is the total number of rows in the final merged dataset.

Saffir-Sampson: $O(1)$

This function uses a control flow of if-else conditions to determine the category of the hurricane based on the wind speed in knots (kts). Every check is a constant-time operation.

Haversine Distance Calculation: $O(n)$

The Haversine function is used in a nested loop. From each county in Florida, the distance is calculated compared to the Hurricane's path. So the operation compared to only three hurricanes becomes linear.

Hurricane Proximity Calculation: $O(n)$

Since this function uses the Haversine and Saffir-Sampson functions, this function also has linear time complexity, since it is a nested loop to compute the proximity between each county and each hurricane.

Normalization: $O(n)$

The normalization function takes $O(n)$, where n is the number of counties that are included in the normalization function. For each county, it needs a normalized value for its energy capacity data, hurricane proximity data, population data, hurricane risk data, and national risk index data. All of these parameters need to be normalized in preparation for the weighted scoring process.

Covariance Matrix Calculation: $O(n)$

The covariance matrix calculation takes $O(n)$, where n is the number of counties with the normalized values. For each county, the normalized values for every parameter are compared to every county, so with n rows and n columns, the overall complexity is the product to produce a matrix $O(n^2)$. This covariance matrix is then used to find the eigenvectors of every parameter to find their weighted scores.

Eigenvector Calculation: $O(n)$

The eigenvector calculation only takes $O(n)$ since the eigenvectors are taken from the covariance matrix, and the eigenvectors associated with the maximum eigenvalues are used to calculate the weights.

Weighted Score Calculation: $O(n)$

The weighted score calculation takes $O(n)$ for every county involved. After all the weights for the parameters are normalized (in other words, after the eigenvector with the largest eigenvalues is normalized), they are added to a total risk score. This risk score determines what counties are the most at-risk during the event of a hurricane, compared to others.

Sorting: $O(n \log n)$

The sorting function takes $O(n \log n)$ based on the number of counties involved in the process. It uses Timsort, from the `sort_values` method in the pandas library which is a combination of insertion sort and merge sort. The overall time complexity is $O(n \log n)$. The counties are sorted by their final risk scores. Each county, with its set of normalized parameters, are sorted to create a heatmap for the most at-risk counties during a given hurricane. The highest risk scores approach 1.0, while the lowest ones approach 0.

Overall, our algorithm takes $O(n \log n)$ based on the number of counties included. Other parameters such as energy capacity are not very big in terms of size, which do not contribute heavily to the total runtime. The worst-case scenario for runtime would be $O(n^2)$, when the number of counties is very large, which requires more normalization and weighing than when the number of counties is smaller. Hypothetically, if a hurricane only impacts two counties, calculating the risk score will be much quicker. If a hurricane's path goes through more than ten counties, the runtime will be a lot slower when our algorithm needs to normalize all values from all registered counties.

VII. Conclusion

In conclusion, our algorithm is correct in detecting hurricane high-risk areas in Florida compared to online hurricane risk tracker tools like Florida's Know Your Zone map and FEMA's

National Risk Index map. Our algorithm's results make sense logically when applied to real-life situations. Counties that are closer to the coastline and are densely populated are more at risk than ones inland and lightly populated. What makes our algorithm unique is its ability to find these risks in real-time. Our implementation implies the path of a hurricane needs to be considered to find the most at-risk areas. A predicted path dataset large enough with the associated category, counties and direction (using coordinates) could be used to find the risk scores of a brand-new hurricane. If the predicted path dataset is too small, it will not accurately predict the most at-risk areas (as it will be too early to tell where the hurricane is going). If we can use the predicted path of an upcoming hurricane, we can find the areas that will be affected the most before the hurricane reaches land! This would allow people precious time to prepare for the storm, provide emergency supplies, send out warnings, and evacuate before the storm reaches the areas. Using a normalization formula, we can see which counties would need warnings or help first and make decisions based on the associated risk scores. While our algorithm's runtime severely depends on how much hurricane path data is available, it has great potential to help others in the event of a hurricane.